



Reproduction in Plants



LEARNING GOALS

- Modes of Reproduction
- Asexual Reproduction
- Sexual Reproduction
- Fruits and Seed Formation
- Seed Dispersal



INSTANT LEARNING

By the end of this chapter, students will understand the concept of reproduction in plants and organisms. They also learn about the various ways of reproduction. Additionally, students will learn about the importance of dispersal in plants.



All living organisms possess a distinctive characteristic to produce their own kind. The production of new individuals from their parents is known as **reproduction**. Now, you must be thinking can plants reproduce?

In this chapter we will learn about the different modes of reproduction in plants.

MODES OF REPRODUCTION

In Class VI, we studied about a flowering plant and its different parts. The root, stem and leaves of a plant are called the **vegetative parts**. Most plants start bearing flowers after a certain period of growth. In spring season, the mango trees can be seen with flowers which further grow into juicy mango fruit that ripens in summer season. While eating fruits, we often throw their seeds. But, a new plant is produced by germination of these seeds. Now, the question arises what is the role of flowers in plants? Flowers are the **reproductive parts** of a plant and their function is to enable reproduction in plants.

FIND OUT

Enlist the various parts of a plant and their functionalities.

A plant can produce its offspring in broadly two ways. These can be classified as:

(i) asexual reproduction (ii) sexual reproduction

In **asexual reproduction**, offspring of a plant is produced without seeds, whereas, in **sexual reproduction**, offspring of a plant is produced from seeds.

ASEXUAL REPRODUCTION

In asexual reproduction, new plants are produced without the seeds through various methods. Let us study about them.

Vegetative Propagation

It is one of the methods of asexual reproduction in which vegetative parts of the plant i.e. the roots, stems, leaves and buds are used to produce the new plants. Thus, this type of reproduction is named as vegetative propagation.

Paheli Inquires

How the plants of sugarcane, potato and rose reproduce? I have never seen their seeds?

ACTIVITY-1

Take a flower of rose or *champa*. Cut its branch along with a node and bury it in the soil. This piece of branch is termed as **cutting**. A part of the stem/branch at which a leaf grows is known as a **node** (Fig. 8.1). Water the cutting regularly and observe it while growing. Observe and note the number of days taken by roots and leaves to grow.

Repeat the same activity by growing money plant in a narrow container filled with water and record your observations.

EXPERIENTIAL LEARNING



Fig. 8.1 Stem-cutting of rose

Flowers grow from the flower buds. In addition to this, the vegetative buds are the another type of bud (Fig. 8.2) These are present in the axil of leaves which grow into shoots. A bud consists of a short stem which is surrounded by immature overlapping leaves. Vegetative buds are capable of growing into new plants.



Fig. 8.2 Potato plant sprouting from an 'eye'

ACTIVITY-2

Take a fresh potato and with the help of a magnifying glass observe its marks. These marks are called as "**eyes**". You will see bud(s) in them. Cut the potato into pieces, each with an eye and bury them in the soil. For a few days water the pieces regularly and observe their growth. What do you observe? Did you notice any change?

In the same way (Fig. 8.3) ginger or turmeric can also be grown.

EXPERIENTIAL LEARNING



Fig. 8.3 Ginger with new plants sprouting from it

Bryophyllum is a sprout leaf plant. It has buds in the its lining of its leaves (Fig. 8.4). If its leaf falls on a moist soil, each bud can produce a new plant.

Likewise, roots of some more plants can also grow into new plants. For example; sweet potato and dahlia.



Fig. 8.4 Leaf of Bryophyllum with buds in the margin

Plants like cacti can initiate the growth of new plants when a part from the main plant gets detached. Each detached part is able to grow into a new plant.

Plants grown by using the technique of vegetative propagation take less time to nurture, to bear flowers and grow fruits as compared to those produced from seeds. As these plants are produced from a single parent, they are the replica of their parent plant.

Further you will learn how plants produced by sexual reproduction comprise features of both the parents and produce seeds as a result.

Budding

We know that tiny organisms such as yeast, which is a single-celled organism are visible only under a microscope and cannot be seen by naked eyes. Such organisms grow and multiply in every few hours if, adequate nutrients are accessible to them by their host. But, how do they reproduce? Let's find out.

Paheli Inquires

What are the advantages of vegetative propagation?



ACTIVITY-3

(To be demonstrated by the teacher)

Buy a piece of yeast cake or yeast powder from a bakery or a chemist shop. Take a container filled with some water. Add a pinch of yeast and a spoonful of sugar into it. Stir it well to dissolve them and let it rest in some warm place. After an hour, place a drop of this liquid on a glass slide and observe it under a microscope. What do you observe? The formation of new yeast cells will be noticed (Fig. 8.5).

EXPERIENTIAL LEARNING

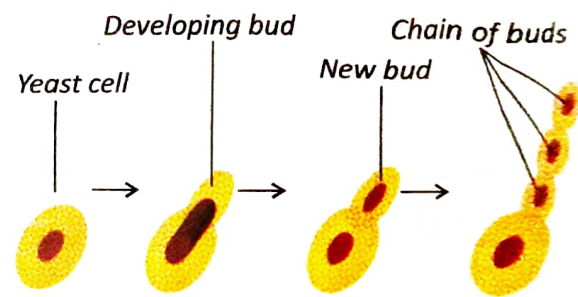


Fig. 8.5 Reproduction in yeast by budding

The small bulb-like projection arising from the yeast cell is called a **bud**. It grows slowly and gets detached from its parent cell resulting in the formation of a new yeast cell. Further, this new cell grows, matures and generates more yeast cells. At times, a chain of buds is formed when another bud comes out from the bud. If the process last for longer duration, then within a short period of time a large number of yeast cells are produced.

Fragmentation

Sometimes, you will see slimy green patches in ponds, lakes or in other stagnant water bodies. What are these? These are the **algae**. Algae grow and multiply rapidly through the process known as **fragmentation**, in the presence of water and nutrients. Algae, in plural is called alga which breaks up into two or more fragments and develops into new individuals (Fig. 8.6). If the process continues, a large area gets covered in a short time span by them.

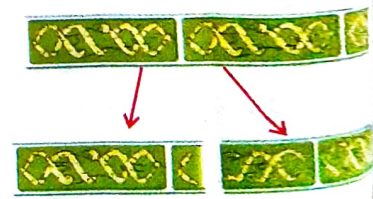


Fig. 8.6 Fragmentation in spirogyra (an alga)

Spore Formation

In Chapter 1, we studied about how fungi grows from the spores which are present in the air. Repeat the Activity 2 of Chapter 1. In the cotton-like lattice on the bread observe the spores. The spores keep floating in the air when released and are able to travel long distances because of their light weight.

Spores are asexual reproductive bodies. In order to tolerate the unfavourable conditions such as high temperature and low humidity, a hard protective coat surrounds a spore which assures its survival for longer period of time. Under favourable conditions, a spore germinates and produces a new individual. The example of plants which reproduce by the means of spores are moss and ferns (Fig. 8.8).

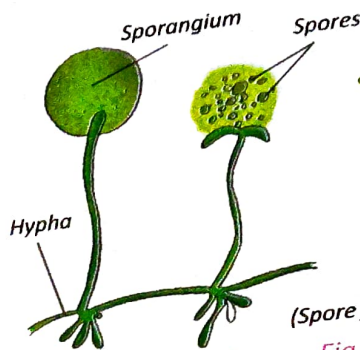


Fig. 8.7 Reproduction through spore formation in fungus

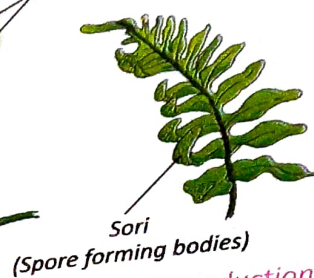


Fig. 8.8 Reproduction through spore formation in fern

SEXUAL REPRODUCTION

Recall the structure of a flower. They are regarded as the reproductive parts of a plant. The male reproductive part is called **stamens** and the female reproductive part is called **pistil** (Fig. 8.9).



ACTIVITY-4

Take a flower of mustard/China rose/petunia and isolate its reproductive parts. Now, study the various parts of a stamen and pistil.

EXPERIENTIAL LEARNING

Flowers which contain only one of the two reproductive part i.e. either pistil or stamens are known as **unisexual flowers**. Whereas, flowers which contain both the reproductive parts are called **bisexual flowers**. Some examples that produce unisexual flowers are corn, papaya and cucumber. On the other hand, mustard, rose and petunia have bisexual flowers. Both male and female unisexual flowers may be present in the same plant or in different plants.

Look at the anther and the filament of a stamen [Fig. 8.9 (a)]. Pollen grains are stored in anther of the stamen which produce **male gametes**. A pistil comprises of stigma, style and ovary. Ovary contains one or more ovules. In an ovule, the formation of **female gamete** or the **egg** takes place [Fig. 8.9 (b)]. A male and a female gamete fuse to form a **zygote**, in sexual reproduction.

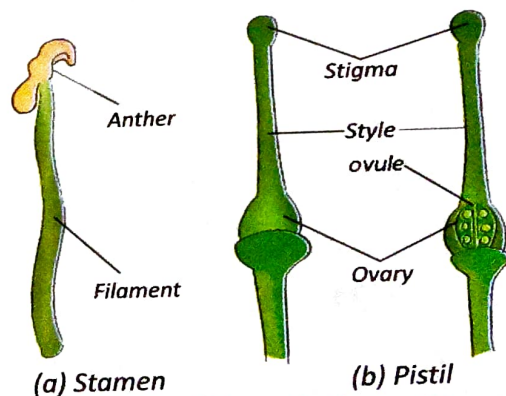


Fig. 8.9 Reproductive parts

Booiho Inquires

How does the male gamete in the pollen grain reaches the female gamete present in the ovule?



Pollination

A tough protective coat prevents pollen grains from getting dried. Pollen grains are easily carried away by wind and water because of their light weight.

Boojo Inquires

Why flowers are so colourful and spreads fragrance? Is it to attract insects?

When insects sit on flowers, pollen grains get stick to their bodies. Some of the pollen falls on the stigma of a flower of the similar kind. The process of relocation of pollen from the anther to the stigma of a flower is known as **pollination**. If the pollen falls on the stigma of the same or another flower of the same plant, then it is called **self-pollination**. Whereas, if the pollen of a flower falls on the stigma of a similar type of a flower but on a different plant then, it is known as **cross-pollination** [Fig. 8.10 (a) and (b)].

Fertilisation

The cell formed after fusion of the gametes is known as **zygote**. When male and female gametes fuse together to form a zygote, then such process is called as **fertilisation** (Fig. 8.11). And, the zygote further develops into an **embryo**.

FRUITS AND SEED FORMATION

The ovary develops into a fruit and other parts of the flower fall off after fertilisation. The fruit is nothing but a ripened ovary. The ovules develop into seeds. An embryo present inside the seed is enclosed in a protective seed coat.

Some fruits are plump and juicy such as mango, apple, orange, etc. while, some are hard like almonds and walnuts [Fig. 8.12 (a) and (b)].

SEED DISPERSAL

You must have observed, that similar kind of plants grow at different places. It usually happens because seeds are scattered to different places. If you have visited a forest, a dense park or a garden then you must have noticed that some seeds or fruits stick to your clothes, after walking through them. Have you ever observed the way in which these seeds were clinging to your clothes?

What will happen if all the seeds of a plant fall at the same place? Obviously, there would be a tough competition among plants to acquire sunlight, water, minerals and space. As a result, the seeds will

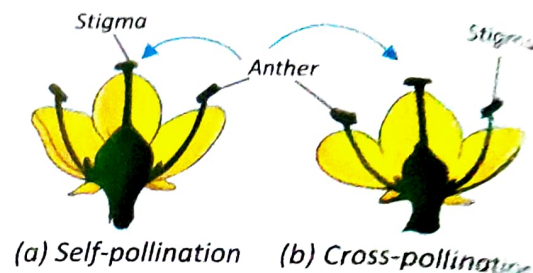


Fig. 8.10 Pollination in flower

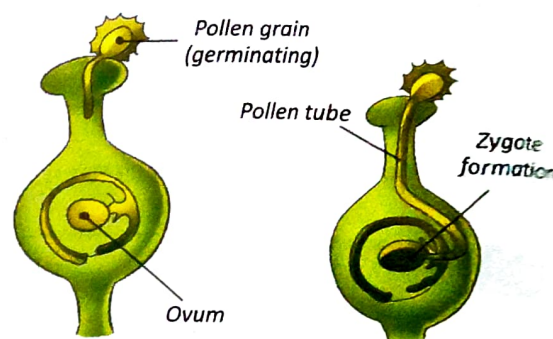


Fig. 8.11 Fertilisation (Zygote formation)



Fig. 8.12 (a) Section of an apple, (b) Almond

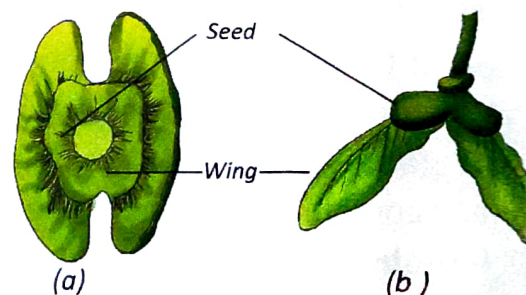


Fig. 8.13 Seeds of (a) drumstick and (b) maple

not develop into healthy plants. Thus, dispersal of seeds is important for plants. This distribution of seeds restricts competition among the plant and its own seedlings for acquiring sunlight, water and minerals. It also enables the plants to seize new habitats for wider distribution.

Wind, water and animals are the few other carriers of seeds and fruits of plants. Winged seeds of drumstick and maple [Fig. 8.13 (a) and (b)], light seeds of grasses or hairy seeds of aak (*madar*) and hairy fruit of sunflower [Fig. 8.14 (a) and (b)], get blown off with the wind to distinct locations. Flow of water also carries away seeds away seeds along with it.



Fig. 8.14 (a) The hairy fruit of sunflower and (b) hairy seed of madar (aak)

Floating ability gets developed in such fruits and seeds in the form of spongy or fibrous outer coat as in coconut. Animals also act as a seed carriers when spiny seeds with hooks get attached to their bodies and is carried to distant places. *Xanthium* (Fig. 8.15) and *urena* are the examples of such seeds.



Fig. 8.15 *Xanthium*

Some seeds also get dispersed when the fruits burst out with sudden jerks. Their seeds get scatters to far-off places from that of the parent plant. For example, castor and balsam.

LET US SUMMARIZE

- The production of new individuals from their parents is known as reproduction.
- The root, stem and leaves of a plant are called the vegetative parts.
- A plant can produce its offspring through asexual and sexual reproduction.
- In asexual reproduction, new plants are produced without the seeds.
- *Bryophyllum* has buds in the lining of its leaves.
- Organisms like yeast grow and multiply every few hours if, adequate nutrients are accessible to them by their host.
- Algae grow and multiply rapidly through the process known as fragmentation.
- A male and a female gamete fuse to form a zygote, in sexual reproduction.
- The dispersal of seeds is important for plants.

EXPLORING WORDS

Asexual reproduction : new plants are produced without seeds

Sexual reproduction : reproduction that involves fusion of male and female gamete

Axil : a point at which leaf is attached to the node

Budding : a mode of reproduction in which a small outgrowth appear on the body of organism

Germination : the process by which a seed gives rise to a new plant

Gametes : the male and female reproductive cells in a flower

Fertilisation : fusion of male gamete with female gamete

Pollination : the transfer of pollen grains from another to the stigma of the same flower

Fruit : the ripening ovary of a plant

Habitat : natural environment in which a particular species of organism live

A. Tick (✓) the correct answer.

- The reproductive part of a plant is the:
 - leaf
 - stem
 - root
 - flower
- The process of fusion of the male and the female gametes is called:
 - fertilization
 - pollination
 - reproduction
 - seed formation
- Mature ovary forms the:
 - seed
 - stamen
 - pistil
 - fruit
- A spore producing organism is:
 - rose
 - bread mould
 - potato
 - ginger
- Bryophyllum can reproduce by its:
 - stem
 - leaves
 - roots
 - flower

B. Fill in the blanks.

- Production of new individuals from the vegetative part of parent is called _____.
- A flower may have either male or female reproductive parts. Such a flower is called _____.
- The transfer of pollen grains from the anther to the stigma of the same or of another flower of the same kind is known as _____.
- The fusion of male and female gametes is termed as _____.
- Seed dispersal takes place by means of _____, _____ and _____.

C. Match items in Column I with those in Column II.

- | Column I | Column II |
|------------------|----------------|
| 1. Bud | a. Maple |
| 2. Eyes | b. Spirogyra |
| 3. Fragmentation | c. Yeast |
| 4. Wings | d. Bread mould |
| 5. Spores | e. Potato |
| | f. Rose |

D. Answer the following questions.

- Describe the different methods of asexual reproduction. Give examples.
- Explain what you understand by sexual reproduction.

3. State the main difference between asexual and sexual reproduction.
4. Sketch the reproductive parts of a flower.
5. Explain the difference between self-pollination and cross-pollination.
6. How does the process of fertilisation take place in flowers?
7. Describe the various ways by which seeds are dispersed.

EXTENDED LEARNING-ACTIVITIES AND PROJECTS

1. Make your own cactus garden by collecting pieces cut from different kinds of cacti. Grow the variety in one single flat container or in separate pots.
2. Visit a fruit market and collect as many local fruits as possible. If many fruits are not available, you can collect tomatoes and cucumbers (these are fruits, though we use them as vegetables). Make drawings of the different fruits. Split the fruits and examine the seeds within. Look for any special characteristics in the fruits and their seeds.

You can visit a library also to learn about this.

3. Think of ten different fruit-bearing plants. Remember that many vegetables are also fruits of the plants. Discuss with your teacher, parents, farmers, fruit growers and agricultural experts (if available nearby) and find out the manner of their dispersal. Present your data in the form of a table as shown below:

S. No.	Name of fruit-bearing plant	Agent through which seeds are dispersed	Part of or seed which helps in dispersal
1.			
2.			
3.			

4. Suppose there is one member of a particular kind of organism in a culture dish, which doubles itself in one hour through asexual reproduction. Work out the number of members of that kind of organism which will be present in the culture dish after ten hours. Such a colony of individuals arising from one parent is called a "clone".



CHAPTER REVIEW

OBJECTIVE TYPE QUESTIONS

UNDERSTANDING SKILLS

A. Tick (✓) the correct option.

1. Which of the following is. The reproductive part of a plant?

a. Leaf b. Stem c. Root d. Flower

